Validity Analysis: MILU

Target context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Overall risk: **MEDIUM**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking teacher in India uses an LLM to evaluate student responses in an examination setting, assign scores with the support of an AI system, and provide feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation at a subsequent stage.

Target population: A postgraduate- or PhD-qualified teacher with 0–20+ years of teaching experience in Hindi, based in North India and interacting in the Hindi language.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	3	Moderate gaps	medium
Input Content ⚠	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ✓	4	Minor gaps	high
Output Content ⚠	2	Significant gaps	medium
Output Form ✓	5	Strong alignment	high
Average	**3.3**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU presents a structurally compatible MCQ benchmark for the North Indian Hindi-medium teacher deployment — the format, output schema, scoring metric, and Devanagari script all match the deployment exactly, and Hindi is the highest-resource Indic language in MILU's scope. However, the HIGH-priority Input Content dimension reveals a critical empirical concern: 100% of sampled Hindi validation items are machine-translated from English (is_translated=True), far exceeding the dataset-wide 25% translation figure cited in the paper. Translated items use English transliterations for technical vocabulary that may not align with state-board Hindi conventions, the only sampled Literature item is about Wordsworth (not canonical Hindi authors), and only one item explicitly references North Indian state content. Combined with absent annotator demographics and no published per-state Hindi-source distribution, the benchmark provides weaker evidence than its India-first framing suggests for the specific construct of evaluating Hindi-medium board-aligned MCQ comprehension. Output ontology, output form, and input form are well-aligned; input ontology is partially aligned with notable gaps in school-literature framing and state civics; input content and output content are the dimensions where validity is most attenuated.

Practical Guidance

What This Benchmark Measures

MILU validly measures an LLM's ability to answer single-correct-answer Hindi MCQs sourced from (or translated to look like) Indian competitive-exam content across 8 broad domains, with reliable binary-scored output that matches the deployment's positive/negative marking schema. The strongest dimensions — Output Form (5), Output Ontology (4), and Input Form (4) — confirm that the benchmark's scoring infrastructure and Hindi/Devanagari modality are a direct fit for this deployment.

Construct Depth

Construct depth is shallow-to-moderate for the deployment's actual target construct (Hindi-medium board-aligned MCQ evaluation). The benchmark probes pan-India competitive-exam-style knowledge in formal Khari Boli, but the dataset analysis reveals that the Hindi validation split is dominated by GPT-4O translations from English source items, which means MILU primarily tests translated-English knowledge rendered in Hindi rather than natively-sourced Hindi-medium curricular knowledge. Coverage of canonical Hindi-board literature (Tulsidas, Premchand), state-specific civics (UP/MP/Rajasthan/Bihar Panchayati Raj), and school-register content (vs. competitive-GK register) is empirically weak in the sampled data.

What Else You Need

To complete the assessment for this deployment, supplement MILU with: (1) a curated Hindi-board test set using items from UP/MP/Rajasthan/Bihar/CBSE Hindi-medium board exams (addresses Input Content and Input Ontology gaps); (2) a Hindi literature MCQ set covering canonical authors in school-literature register (addresses the Wordsworth-vs-Tulsidas Arts & Humanities gap); (3) a state civics and Panchayati Raj MCQ set (addresses Input Ontology under-representation); (4) Hindi-medium teacher annotator validation of a sample of translated MILU items to estimate translation-induced answer-key error rate (addresses Output Content and Input Content concerns); (5) verification that production inference mode matches the evaluation method used (log-likelihood vs. generative) per [Q85].

ID	Evidence
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)